

1. Administrative & Study Identification

Full Study Title: Use and Effectiveness of COVID-19 Vaccines Using State Vaccine Registries and Insurance Claims Data **Short Title/Acronym:** Pfizer COVID-19 Vaccine RWE Study
Protocol Number: [Internal Study ID]
NCT Number: NCT05876377 **Sponsor Name:** Pfizer **Investigational Product:** Pfizer-BioNTech COVID-19 Vaccine (Original and Bivalent formulations) **Phase of Development:** N/A (Observational/Real-World Evidence)
Medical Monitor: Study Director, Pfizer CT.gov Call Center

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To measure how effective the Pfizer-BioNTech vaccine is at preventing post-COVID conditions (Long COVID). **Secondary Objectives:** To understand the characteristics (e.g., demographics, comorbidities) of patients who are receiving COVID-19 vaccines.

2.2 Background & Rationale To address the gap in real-world evidence regarding the long-term effectiveness of the Pfizer-BioNTech COVID-19 vaccine in mitigating post-acute sequelae of SARS-CoV-2 infection. Using vaccination information from state registries linked with HealthVerity insurance claims provides critical insights into vaccine outcomes and adoption across diverse populations.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Retrospective/Prospective Claims Database Cohort Study) **Control Type:** Unvaccinated cohort or alternate formulation comparator **Blinding:** Open-label / Unblinded **Randomization:** N/A (Observational Database Study)

3.2 Planned Enrollment & Duration Target **\$N\$:** Evaluated based on all eligible patient records residing in the HealthVerity database (primarily capturing California and Louisiana). **Number of Sites:** N/A (Database query) **Estimated Study Start:** August 31, 2022 (Index eligibility date) **Estimated Primary Completion:** August 26, 2025 (End of observation window for post-COVID diagnosis)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Patients must have medical and pharmacy enrollment in HealthVerity (with at least 6 months of continuous enrollment before the calendar index date). 2. Age ≥ 6 months (with subgroups evaluated across various age ranges, e.g., 5-11, 12-49 pregnant women, adults ≥ 18). 3. Patients are required to have a COVID-19 diagnosis on or after September 5, 2024, and no later than August 26, 2025 (specifically residing in California or Louisiana for Aim 1). 4. Evidence of ≥ 2 or ≥ 3 doses of the COVID-19 original vaccine formulation before the calendar index date (depending on the specific cohort).

4.2 Exclusion Criteria 1. Received the original vaccine formulation within 2 months before the calendar index date. 2. For specific cohorts: Evidence of moderate/severe immunocompromise as defined by the CDC. 3. Did not maintain continuous medical/pharmacy enrollment for ≥ 1 year prior to the cohort entry date (allowing up to a 30-day gap).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: ≥ 2 or ≥ 3 doses of the Pfizer-BioNTech COVID-19 Vaccine (inclusive of original and bivalent formulations). **Route of Administration:** Intramuscular (IM) injection (documented via vaccine registries). **Duration of Treatment:** N/A (Observation of single or series administration followed by longitudinal claims follow-up).

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care for COVID-19 infection and underlying co-morbidities. **Prohibited:** N/A (Real-world observational tracking).

6. Study Timeline & Visit Schedule

(Note: As this is a claims-based observational study, actual physical visits are substituted with database queries representing specific clinical milestones)

Visit ID	Window	Key Activities/Procedures
Screening	Day -180 to 0	Database extraction: Confirm HealthVerity enrollment, demographics, and prior vaccine history.
Baseline	Day 0	Index Date (Date of confirmed COVID-19 diagnosis or vaccination milestone).
Interim	Post-Index	Longitudinal tracking of insurance claims for clinical symptoms and diagnoses.
End of Study	Up to Aug 2025	Final data extraction for post-COVID condition diagnosis incidence.

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Effectiveness of the vaccine at preventing post-COVID conditions (Long COVID). **Measurement Method:** ICD-10 diagnostic codes and pharmacy claims queried from the HealthVerity database.

7.2 Secondary & Exploratory Endpoints Characteristics of patients receiving COVID-19 vaccines (evaluated through demographic segmentation, geography, and pre-existing medical claims).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: N/A (Standard prospective clinical AE collection is not applicable for retrospective/prospective claims analysis). **Serious Adverse Events (SAEs):** N/A. **Data Monitoring Committee (DMC):** Pfizer internal review and epidemiologic quality control teams.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Patients residing in California and Louisiana within the HealthVerity database who meet the medical/pharmacy enrollment criteria. **Sample Size Justification:** Determined by the natural incidence of COVID-19 diagnoses and vaccination rates within the queried database period rather than traditional power calculations. **Handling of Missing Data:** Standard epidemiologic methodology for real-world evidence (e.g., excluding patients with >30 day gaps in insurance coverage).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: Conducted in accordance with Good Pharmacoepidemiology Practices (GPP) and health data privacy regulations (HIPAA compliance for de-identified data). **Monitoring Plan:** Continuous data validation, claims abstraction tracking, and statistical programming code review. **Audit Trail:** Logs of HealthVerity database queries and analytic derivations.

11. Study Identifiers & Governance

Primary ID: [Internal Study ID] **Registry Number:** NCT05876377
Sponsor/Lead Organization: Pfizer **Study Title:** Use and Effectiveness of COVID-19 Vaccines Using State Vaccine Registries and Insurance Claims Data **Official Regulatory Title:** Use and Effectiveness of COVID-19 Vaccines Using State Vaccine Registries and Insurance Claims Data

12. Clinical Framework (*Adapted for COVID-19*)

Primary Condition: COVID-19 / Post-Acute COVID-19 Syndrome
Diagnosis Threshold: Documented COVID-19 or Long COVID diagnosis (ICD-10) via insurance claims **Medical Methodology:** COVID-19 Vaccination (Pfizer-BioNTech) **Optimization Target:** Prevention of post-COVID conditions

13. Study Procedures & Methodology

Management Pathway: Real-world observation of state vaccination registries and HealthVerity claims **Education Protocol (HCP):** N/A (Observational Database Study) **Education Protocol (Patient):** N/A (Observational Database Study) **Quality Audit Mechanism:** Statistical validation of health claims and vaccine registry linkage

14. Observation & Follow-Up Schedule

Total Duration: Retrospective and prospective claims observation (up through August 2025) **Onsite Visit Cadence:** N/A (Database extraction) **Remote Monitoring Frequency:** Routine database sweeps/linkages **Checklist Review Interval:** Continuous programmatic analysis

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator / Study Director											[Name]		_____		[DD- MMM-YYYY]	
	Clinical Operations Lead											[Name]		_____		[DD- MMM- YYYY]	
	Lead Statistician											[Name]		_____		[DD- MMM- YYYY]	